

FACIAL AND GAIT CUES FOR MENTAL HEALTH DETECTION

Submitted in partial fulfillment of the requirements for the award of Bachelor of Engineering degree in Computer Science and Engineering with Specialization in Artificial Intelligence and Machine Learning

By

GIDUTURI SIRI JYOTHI (Reg. No-42611138)

KUNCHALA DEEPTHI (Reg. No- 42611179)



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
SCHOOL OF COMPUTING**

SATHYABAMA

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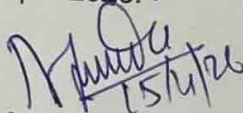
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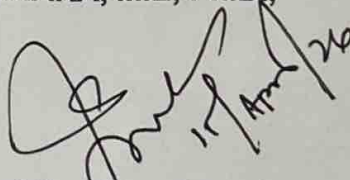
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BONAFIDE CERTIFICATE

This is to certify that this Project Report is the bonafide work of **GIDUTURI SIRI JYOYHI (42611138), KUNCHALA DEEPTHI (42611179)**, who carried out the Project entitled **"FACIAL AND GAIT CUES FOR MENATL HEALTH DETECTION"** under my supervision from November 2025 to April 2026.


Internal Guide

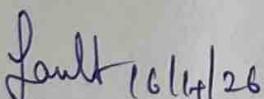
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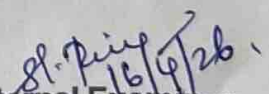


Head of the Department

Dr. SENDURU SRINIVASULU, M.Tech., Ph.D.,

Submitted for Viva voce Examination held on 16/04/2026


Internal Examiner


External Examiner

DECLARATION

I, **GIDUTURI SIRI JYOTHI** (Reg. No- 42611138), here by declare that the Project Report entitled "**FACIAL AND GAIT CUES FOR MENTAL HEALTH DETECTION**" done by me under the guidance of **Dr. D. MENAKA, M.E, Ph.D.**, is submitted in partial fulfillment of the requirements for the award of Bachelor of Engineering degree in **Computer Science and Engineering with Artificial Intelligence and Machine Learning**.

DATE: 16/04/2026

G. Siri Jyothi

PLACE: Chennai

SIGNATURE OF THE CANDIDATE

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ABSTRACT

There is little to no accessible and cost-efficient methods for screening depression that can be scaled and implemented to the over 280 million individuals suffering from Major depressive disorder worldwide. We created a prototype system that assesses depression risk by analyzing facial expressions and walking patterns from standard videos. Using a ResNet-18 architecture, we derived 512-dimensional facial feature representations from facial portraits and processed Media Pipe 17-joint gait representations (51 features/frame) with a bidirectional LSTM that captures movement-invariant temporal patterns. The resulting feature sets from each classifier are input to a small early-fusion multilayer perceptron (MLP) to produce a binary Healthy vs. Depressed classification. Initial assessment of model performance with 12 video-face pair tests showed that the models behaved differently. The gait model, for example, showed a significant bias towards Healthy classifications (mean confidence $\approx 69\%$), and the fusion model outputs were concentrated near the threshold of the decision boundary. Such we see to be shifts in distribution between what we put into the training sequences and what is present in natural settings which in turn plays out in landmark extraction accuracy and motion variability. We present that it is of great importance for preprocessing and for the expansion and balance of data sets. Also the framework we present proves out that technical viability of multi-modal analysis on smart phones is a reality which in turn we are using as a base to develop depression screening applications which are production grade.

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CHAPTER 1

INTRODUCTION

Depression is a very present issue which we see to be the most common of mental health issues which affect people across age and cultural groups worldwide [1]. While we have seen an increase in awareness in recent years identification of the disease at the early stages is still a challenge in many real world settings. At present most clinical tools we have are questionnaires or interview based such as the PHQ-9 which ask individuals to report on their mental state [2]. In the real world we see that symptoms are reported under and also very ambiguously which is especially true for early and mild stages of the disorder. To get around these issues research has been directed toward objective methods which look at behavior which is measurable. Human movement and facial behavior in particular are of interest as they may be passively captured via video without disruption of daily life.

In the past we reported that which is to say that we have seen that depression goes along with visible changes in gait which include decreased speed, less variability, and postural changes [3], [4]. Also we report that we have which which we note that gait based models for depression which we train on video or depth data do perform well but also that they are very much at the at the mercy of the recording conditions and also the characteristics of the used dataset [5]. Also we see that facial expressions play a large role in terms of which we report that which is which which in fact what we have is that people with depression present with reduced facial expression and different emotional patterns [6]. With the growth of deep learning which we see is which which in turn has brought about very which improved ability of facial analysis systems to put out which we may term as the very fine scale affective information from images and video [7], [8].

To address the issues of unimodal systems' performance, multimodal approaches which have seen great growth in recent times do so. We see that which put forward a combination of facial expressions, gait patterns and other behavioral elements out to present a more complete picture of depression related behavior [9], [10]. Also we find from research that these models do perform better than their unimodal counterparts in practice [11]. But also with this comes issues of data alignment, class imbalance and design of fusion strategies.

Recent we have seen that there has been great improvement in gait analysis from standard RGB video which is a result of recent advances in pose estimation. Frameworks like Media Pipe Pose which do away with the need for special sensors thus making large scale and smartphone based deployment a reality [13], [14]. By way of these improvements we looked at a multi modal framework which puts together facial expression analysis with gait features from everyday video[15]. Instead of reporting only on peak accuracy we looked at what the models do, the prediction biases, and the practical issues when applied to real world data which is done with the aim of putting in place reliable and clinical grade videobased depression screening systems.

Beyond the technical advancements, an important aspect of multimodal depression detection lies in its potential for real-world deployment and accessibility. Unlike traditional clinical assessments that require active participation, video-based analysis allows for passive and continuous monitoring[16], [17]. This is particularly beneficial in cases where individuals may be unwilling or unable to articulate their emotional state. For example, early-stage depression often manifests subtly, with small behavioral cues such as reduced eye contact, slower movements, or diminished expressiveness signals that may go unnoticed in self-reported assessments but can be captured through computational models[18]. real-time market intelligence, it enables precise, personalized, and adaptive career guidance.

CHAPTER 2

LITERATURE SURVEY

Stolar et al. (2023)[1] states that People with depression show diminished facial action unit (AU) activations related to smiling, especially those related to positive emotions, including AU12, lip corner puller. Diminished facial expressiveness is a part of a more general problem with positive emotions, as people with depression show blunted or less frequent smiling, which aligns with the observation that depression is associated with diminished reactivity to positive stimuli and the intensity of happy facial expressions. Moreover, diminished smiling-related AUs go hand-in-hand with difficulties in recognizing positive emotions in others, which points to a problem with emotional perception as well as expression

Islam et al. (2024)[2] introduced an advanced AI-based system called Emo Pulse that is capable of detecting human mental states by incorporating facial emotion recognition in combination with psychometric assessment information. The Emo Pulse system utilizes a deep learning model called Emo NN that is able to process facial expressions to detect human emotion-related information such as happiness, sadness, anger, and even neutral states. Unlike other conventional approaches that only utilize visual information to detect human emotion states, Emo Pulse is able to improve the accuracy of emotion state detection by incorporating psychometric information. This information is used to obtain a comprehensive view of an individual's mental state in detecting mental health problems such as depression and anxiety disorders.

Sharma et al. (2024)[3] undertook a comprehensive study of facial Action Unit (AU) sequences to understand the relationship between AU sequences and mental health conditions like depression and anxiety. The study highlights that it is not only the individual Action Units that have a significant impact in the identification of underlying emotional states but also the sequence and co-occurrence of multiple Action Units. By using AU sequence models, the researchers were able to successfully identify the subtle changes in facial expressions that are not captured in static images. The study has shown that specific AU sequences characterized by reduced presence of positive

expression units and increased presence of tension-related/negative expression units have a significant correlation with depression and anxiety.

Researchers at Dartmouth (2024)[4] proposed an innovative smartphone-based AI application called "Mood Capture," which can identify depression symptoms based on facial data from the front camera of smartphones. The application collects images of individuals through passive smartphone use and analyzes facial expressions and behavioral patterns through computer vision and deep learning techniques. Unlike other depression detection techniques that use personal reports and interviews, "Mood Capture" offers a novel and non-intrusive method of depression detection. The study showed promising accuracy of about 90% in detecting individuals with depression symptoms through smartphone use.

Islam et al. (2024)[5] proved the superiority of Convolutional Neural Network (CNN)-based deep learning approaches over the conventional handcrafted feature-based approaches for the task of emotion recognition. The conventional approaches are based on the extraction of handcrafted features such as texture features, geometric features, or statistical features, which are not sufficient enough to handle the complex variations in human facial expressions. The CNN-based approaches, on the contrary, are able to learn the feature representation directly from the raw images, making them competent enough to recognize low-level features such as edges and contours, as well as high-level features such as expressions and emotional cues.

The important role of deep facial feature extraction in the improved classification of mental health conditions like depression and anxiety has been pointed out in the study by Sharma et al. (2024)[6]. This study also highlights the ability of deep learning techniques like neural networks to automatically learn the extraction of high-level and discriminative facial features, which are difficult to obtain through other conventional techniques. This helps the model in the detection of the slight changes in facial expressions and other micro-expressions of the human face, which are associated with the underlying mental health conditions.

Liu et al. (2024)[7] presented a multimodal depression detection model by using a combination of gait analysis and standardized clinical rating scales. The research uses the patterns of gait, including the speed of walking, posture, and other dynamics, as a form of behavioral indicator, as well as subjective feedback from rating scales for depression. The combination of the two data sources, which provide a balance between the physical and mental manifestations of depression, enables the model to provide a more holistic approach to depression detection.

Cosma et al. (2024)[8] the authors proposed a novel dataset called Psy Mo, which is intended for the evaluation of certain psychological characteristics by employing gait analysis with the help of camera-based pose estimation approaches. The dataset involves the observation of human walking patterns in a real- world setting, as well as the application of computer vision-based approaches to the observation of the dynamics of human skeletons. The research leverages the capabilities of pose estimation to facilitate the non-invasive observation of certain behavioral characteristics, which are directly related to the psychological state of a person.

Frontiers in Psychiatry (2021)[9] has established a strong link between depression and changes in motor behavior that manifest as reduced head movements, reduced limb movements, and a general slowing down of the gait. Such motor behavior is characteristic of psychomotor slowing, which is one of the symptoms of depression that is characterized by reduced physical and cognitive processing speeds. The study showed that people living with depression tend to have reduced dynamic and more rigid motor behavior characterized by reduced walking speeds, reduced stride length, and reduced variability in motor movements.

Liu et al. (2024)[9] the authors used a combination of Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) to classify the video- based gait patterns for the detection of depression. In the authors' method, CNN models were used to extract the spatial features of the individual video frames, which included the structural information as well as the posture-related information, while the RNN models were used to analyze the temporal dependencies between the consecutive video frames, which helped to analyze the dynamics of the movement.

For instance, Microsoft Kinect technology and pressure-sensitive walkway technology offer effective non-invasive methods of tracking human skeletal gait. The former uses depth sensor technology coupled with infrared imaging to track three-dimensional skeletal joint positions. This facilitates accurate tracking of human body posture and movement without requiring wearable devices. Similarly, pressure-sensitive walkway technology tracks foot pressure distribution, step patterns, and gait time parameters such as stride length and walking speed. These methods facilitate uninterrupted objective tracking of human movement in various environments. The tracking of gait parameters in real time facilitates the identification of behavioral changes in patients suffering from depression.

The GT-CLIP model makes use of the attention-guided multimodal learning framework in the process of improving the feature extraction process. The GT-CLIP model is inspired by contrastive learning techniques, which align the representations of the visual gait and text data in the shared embedding space. The attention mechanism is also important in the process of improving the performance of the model in the process of feature extraction. This is because the attention mechanism helps the model focus on the important parts of the gait data, such as the movement of the joints in the human body.

Martinez-Martin (2023), the ethical issues associated with the use of facial and gait characteristics in AI-based mental health and behavioral analysis systems were considered, with particular focus on the major ethical concerns associated with privacy, consent, and data security. In this study, the major ethical concerns are highlighted with regard to the use of facial and gait characteristics, which are considered to be highly sensitive and can reveal not only the identity of the individuals but also their underlying mental states. This makes the misuse of these characteristics in AI-based mental health and behavioral analysis systems a major ethical concern.

Brookings Institution (2023) has stressed the need for more stringent privacy regulations regarding the usage of AI tools in the field of mental health applications. In the report, it has been mentioned that AI-based technologies make use of highly sensitive personal data, including facial expressions, behavioral patterns, and psychological patterns. This has raised serious concerns regarding the misuse and unauthorized access of the data. It has also been mentioned that the current regulations regarding data privacy are not adequate to address the risks associated with the usage of mental health technologies, especially in the case of continuous monitoring and passive collection of data.

Report from The Lancet in 2023 talks about how biased data in AI could make healthcare problems worse, especially for inequalities that already exist. It seems like if we don't fix this, things might get really uneven. Many AI models get trained on data that's not very diverse, you know, missing out on different ethnicities or genders or ages and stuff like that, plus socioeconomic backgrounds too. That means when these systems try to work with people who were in the training data, they often mess up, like giving wrong predictions or diagnoses.

In the early days of research, a lot of attention was given to analyzing facial expressions as a crucial sign of mental health issues. Various studies have indicated that even the tiniest facial movements, known as micro-expressions, along with eye behavior, can uncover deeper psychological states. For example, investigations into facial movement analysis have shown that these subtle features can help predict a range of psychological symptoms, making them a dependable tool for assessing mental health. Likewise, more recent large-scale studies have pointed out that facial signals—like where someone is looking, how their pupils move, and the dynamics of their expressions—are closely linked to mood disorders such as depression and anxiety.

2.1 Inferences from Literature Survey

Current AI systems for detecting mental health stuff usually just pick one thing, like facial expressions or how someone walks, and they do not really combine them in a full way. That seems like a big problem because emotions and conditions such as depression show up through all sorts of signals, not just one. It is kind of obvious that humans express feelings in a mix of behaviors, body changes, and thoughts. Most of the approaches out there stick to a single modality. Facial cues, speech, or gait patterns, something like that. But relying on only one source means you are missing out on other important parts that could make the picture clearer.

Many models trained on datasets that were not particularly diverse, and their performance dropped when applied to people from different ethnicities, ages or cultural backgrounds. Another main limitation in existing mental health detection systems is poor generalization across diverse populations. However, a lot of AIs use data that is heterogeneous in demographic diversity and thus often under-represent variations among age, gender, ethnicity, and culture. Consequently, these models provide excellent performance on the dedicated data used for training but quite less generalizability to new or unseen populations.

One of the fundamental drawbacks in traditional mental health assessment is the high degree of subjectivity involved with diagnosis. Most existing processes depend on self-reported questionnaires, clinical interviews, and expert evaluations that can vary widely based on a person's perception, readiness to disclose symptoms and the clinician's interpretation. However, if a patient is not able to communicate how they feel due to perceived stigma in society or perhaps not being aware of what those feelings are, then we run the chance that relevant information we may obtain through scales presented to patients may be inconsistent or incorrect. So-called subjective symptoms that are experienced by one clinician may be interpreted differently by another clinician, leading to variability in diagnosis and treatment decisions.

The development and deployment of AI-based mental health detection systems faces major challenges because of privacy and ethical issues. The technologies used in these systems need to gather personal data which includes facial expressions and gait patterns and voice recordings and behavioral information to access private details about a person's identity and mental health. People who gain unauthorized access to private information through data breaches or information misuse will face severe consequences that include discrimination and stigmatization and loss of trust. The process of securing informed consent becomes more complicated when systems use passive or continuous data collection methods because users do not understand how their data will be collected and utilized.

AI-based mental health detection systems experience their second major restriction through their sensitivity to environmental conditions which affects their performance. Many models, especially those relying on visual and video data, experience major disruptions due to external factors which include lighting variations and camera quality and background noise and occlusions and changes in viewing angles. Poor lighting and low-resolution images make it difficult to extract facial features accurately while cluttered backgrounds and partial occlusions block the view of body movements and walking patterns. The different recording environments, which include indoor and outdoor settings, create data inconsistencies that affect the collected information. The system requires users to take responsibility for any mistakes that lead to incorrect results or wrong classifications.

Another important point to consider is that our gait and behavior can reveal a lot, especially when it comes to understanding the physical and motor changes linked to mental health issues. That said, analyzing gait isn't as reliable as looking at facial features, particularly in everyday or chaotic settings. This inconsistency often arises from factors like varying recording conditions, obstructions, different walking styles, and the limitations of preprocessing methods like pose estimation.

The literature emphasizes the increasing significance of deep learning techniques in this field. Approaches like Convolutional Neural Networks (CNNs) and recurrent models have shown to outperform traditional machine learning methods, as they can automatically learn intricate and high-dimensional feature representations. This not only lessens the reliance on manual feature engineering but also enhances overall accuracy and generalization.

Recent research is that multimodal systems greatly surpass unimodal approaches. By integrating facial recognition, gait analysis, and sometimes even additional modalities like speech or physiological signals, these systems can gather complementary information, resulting in better prediction accuracy, robustness, and reliability. This finding strongly advocates for the use of a multimodal framework in the proposed work.

Another crucial insight pertains to model evaluation and reliability. Numerous studies highlight the importance of using appropriate evaluation metrics such as ROC curves, confusion matrices, and calibration analysis. It has been noted that high-performing models are not just accurate but also well-calibrated, meaning they offer trustworthy confidence estimates instead of overly confident predictions. This is especially vital in mental health applications, where incorrect predictions can have serious consequences.

The literature review on Facial and Gait Cues for Mental Health Detection offers some valuable insights that can really shape the design and development of the proposed system. One major takeaway is that facial expressions serve as powerful indicators of mental health issues. Numerous studies have shown that elements like micro-expressions, eye movements, and emotional fluctuations can effectively mirror psychological states such as depression and anxiety. This highlights facial analysis as a solid and trustworthy foundation for detecting mental health conditions.

2.2 Open Problems in Existing System

The current AI mental health detection systems face a major challenge because they depend on small biased datasets which do not provide sufficient data for their operation. Many studies are conducted using limited data samples that do not adequately represent the diversity of real-world populations in terms of age, gender, ethnicity, and cultural background. The datasets create a situation which causes overfitting because models achieve successful results on training data but they cannot perform well with new unseen data. The model learns distorted patterns because biased datasets create learning conditions which lead to different prediction outcomes between demographic groups. The problem becomes crucial for mental health systems because wrong classification results in severe negative effects.

The presence of subjective judgment in ground truth labeling procedures creates a major obstacle that hinders AI-based mental health detection systems from functioning properly. The foundation of ground truth data, which machine learning models use for their training and evaluation process, comes from self-reported questionnaires and clinical assessments and expert annotations. The sources exhibit substantial variability because people have their own distinct ways of perceiving things and understanding their emotions and sharing their health issues. Patients who experience stigma or fail to comprehend their condition will provide incorrect answers through their responses, while clinicians who observe behavior will assess it according to their personal expertise and assessment methods.

When developing AI-based mental health detection systems, privacy and ethical issues are major obstacles. These systems frequently rely on extremely sensitive data, such as voice signals, gait patterns, facial expressions, and behavioral information, which can disclose a person's identity as well as their emotional and psychological state. There are serious risks associated with data breaches, unauthorized access, and possible misuse when such data is collected, stored, and processed.

A major drawback of AI-based mental health detection systems is their sensitivity to environmental factors, especially when they depend on motion-based and visual data like gait patterns and facial expressions. The quality of recorded data and, in turn, the precision of feature extraction can be greatly impacted by variations in lighting, camera resolution, background clutter, occlusions, and viewing angles. For instance, dim lighting can make it difficult to see facial features, and complicated or dynamic backgrounds can make it difficult to accurately estimate a person's pose and track their gait. Furthermore, inconsistencies that lower model robustness are introduced by differences between controlled laboratory settings and real-world environments.

AI-based mental health detection systems that rely on facial and gait analysis face serious difficulties due to occlusion and pose variability. Occlusion results in incomplete or distorted feature extraction when portions of the face or body are partially obscured by objects like glasses, masks, hair, or environmental obstacles. Similarly, it can be challenging for models to maintain consistency in recognition because changes in head orientation, body posture, and camera angles can change the appearance of facial expressions and gait patterns. These elements may cause important features to be missed or detected incorrectly, which would lower the system's overall performance and dependability.

AI-based mental health detection systems face a challenge because cultural and social factors greatly influence how people express and perceive emotions. Behavioral patterns, body language, and facial expressions can differ greatly between cultures, social norms, and situations. For example, some cultures may discourage overt emotional expression, resulting in more subdued or restrained expressions, while others may behave more expressively. Similar to this, social elements like upbringing, surroundings, and interpersonal relationships can affect how people react to emotional circumstances.

One of the big challenges we face is the lack of large and diverse datasets. Most of the studies out there depend on small, controlled datasets that just don't capture the real-world diversity we see in age, gender, ethnicity, and environmental conditions. Because of this, models trained on these limited datasets often struggle to perform well when they encounter new or different populations. This issue of generalization is a significant hurdle when it comes to using these systems in real-life situations.

Another major hurdle is dealing with unstructured and real-world data. In everyday settings, facial expressions can be influenced by things like lighting changes, obstructions, camera angles, and background noise. The same goes for gait analysis, which is really sensitive to factors like uneven surfaces, different clothing, and how the camera is positioned. Unfortunately, many existing systems have a tough time keeping their performance consistent in these uncontrolled environments, which makes them less dependable outside of lab conditions.

The limited ability of gait features to distinguish between different scenarios is still a challenge we face. While gait analysis can offer valuable insights into behavior, its effectiveness often takes a hit when the features we extract are noisy or when the pose estimation isn't quite right. A lot of current methods rely heavily on the quality of preprocessing, and any mistakes in silhouette extraction or joint detection can really hurt performance.

Another significant challenge is the absence of strong multimodal fusion strategies. Even though multimodal systems have demonstrated better performance, effectively merging different types of data is still a tricky endeavor. Many of the methods we have now use basic fusion techniques that don't fully take advantage of the complementary strengths of facial and gait data. We really need more sophisticated fusion mechanisms that can adaptively assess the importance of each modality depending on the context.

Model interpretability and explainability are also major concerns. Most deep learning models operate like "black boxes," giving us predictions without any clear reasoning behind them. In sensitive areas like mental health, it's vital to understand the rationale behind a specific decision. Unfortunately, many existing systems lack transparency, which can undermine trust and make it harder for healthcare professionals to adopt them. Another ongoing issue is model calibration and overconfidence. Many models tend to give overly confident predictions, even when they're wrong, which can be particularly risky in mental health scenarios. Making sure that models provide trustworthy confidence estimates is a challenge that still needs more attention and research.

CHAPTER 3

AIM AND SCOPE OF THE PRESENT INVESTIGATION

From a technical, operational, and practical standpoint, the suggested multimodal depression detection system, which combines gait recognition and facial expression analysis, shows strong viability. Technically speaking, efficient model development for feature extraction and classification is made possible by the availability of sophisticated deep learning frameworks like TensorFlow and Py Torch. Additionally, data acquisition is made accessible and scalable through the use of inexpensive hardware, such as cameras and sensors. Pre-trained models and existing datasets facilitate quick prototyping and shorten development times. From an operational standpoint, the system's non-intrusive and user-friendly design enables ongoing monitoring without necessitating active user involvement. This improves usability in real-world settings like remote monitoring environments, workplaces, and healthcare facilities.

The system becomes more robust and reliable through its multimodal data integration which merges facial signals with walking patterns and potential psychological test results. The system requires immediate attention for its multiple security weaknesses which have been identified. The system processes personal data which contains both private information and behavioral tracking details which need protection against security threats. The system needs to maintain data security through proper storage methods and encryption techniques while following all relevant data protection laws. The training data contains bias which creates challenges for the model to produce fair results when applied to various population groups. The system performance receives influence from environmental elements which include the way light appears and the position of cameras and the amount of sound present in the background. The system requires users to agree through consent procedures while it must demonstrate full visibility about its data processing activities and its methods for understanding user information. The system faces classification risks which produce false positive and false negative results that result in wrong evaluation outcomes.

3.1 Aim of the Present Investigation

The research investigates how scientists apply deep learning systems together with modern artificial intelligence technology to detect vital behavioral and physical characteristics which show what people experience mentally. The system unites various data types to solve problems which single-source methods face when they try to deliver complete and precise results. The system uses facial action unit analysis together with emotional expression detection and body posture evaluation and walking pattern observation to detect small signs which indicate mental health issues. The project investigates methods for combining different data types to achieve better prediction results which remain stable across various situations. The system operates across multiple fields which include healthcare surveillance and clinical starting point testing and employee psychological health evaluation and distant health monitoring systems.

Its focus on developing a non-intrusive monitoring system which will provide affordable solutions that operate with real-time performance in live operational settings. The system requires developers to address three fundamental ethical aspects which include protecting user data privacy and obtaining proper consent from users and creating AI systems that function without any form of discrimination. The system operates as an auxiliary tool which helps doctors identify mental health conditions at early stages while they can monitor patients throughout their treatment process to achieve better mental health results. The evaluation process for the proposed system investigates four essential elements which include technical and operational and economic and ethical components.

From a technical perspective, the project is highly feasible due to the availability of powerful deep learning frameworks such as TensorFlow and Py Torch, which support efficient model development, training, and deployment. The implementation of complex multimodal models becomes more feasible because high-performance computing resources now exist in the form of GPUs together with cloud platforms. The system operates with an interface which stays simple for users while it performs its tasks without bothering them through its use of standard smartphone and camera devices which gather information. The system operates effectively for actual field work because it supports healthcare facilities together with medical practice offices.

3.2 Scope of Present Investigation

The current research focuses on building an intelligent system which combines facial expression analysis with gait pattern recognition to identify early mental health symptoms of depression and its related disorders. The research establishes its boundaries through the combination of computer vision systems and machine learning algorithms which produce a mental health assessment system that works without physical contact and provides automated results. The research focuses on analyzing facial signals which include brief facial expressions and eye tracking and muscle activity in the face and emotional changes that appear in video recordings and photographic images. The research team conducts gait analysis through body posture assessment and walking speed measurement and stride length evaluation and movement pattern analysis which help identify psychological conditions of patients.

The research requires development of deep learning systems which merge CNNs to identify facial characteristics with LSTM models which track movement patterns through time. The system combines different modalities to achieve better detection performance than systems which use individual modalities. The research focuses on structured data sets which scientists collect by using predefined testing environments for facial and gait data acquisition. The system works as a screening tool which helps mental health experts through its supportive features although it does not aim to substitute for human diagnosis. The investigation uses multiple evaluation metrics which include accuracy and precision and recall and F1-score and ROC analysis to assess system performance. The research process evaluates ethical concerns which involve protecting personal data and obtaining proper authorization and identifying discriminatory patterns in dataset information. The system faces potential operational constraints because its performance might fail to match laboratory results when used in actual field situations. The research project focuses on creating an AI-based system which uses multiple data types to identify mental health problems during their initial phase through analysis of face and body movement patterns. The research team aims to develop an AI-based multimodal system which will detect early signs of mental health problems through analysis of facial expressions and walking patterns to improve healthcare systems.

3.3 Software Requirements Specifications

The system receives development through current programming tools which link with deep learning frameworks for its construction. Machine learning and computer vision applications receive their primary programming language from Python which provides extensive support for these fields. The implementation of deep learning models through TensorFlow and Py Torch libraries enables the creation of CNN and LSTM and MLP models. Users can process videos and extract frames through OpenCV while Media Pipe allows them to extract human gait features by estimating body positions. The software performs facial landmark detection through Dlib and Open Face libraries. The users operate NumPy and Pandas for data processing and management while they use Scikit-learn to build their machine learning models based on standard methods. The development environment includes Jupyter Notebook or Visual Studio Code, and the system can run on operating systems such as Windows, Linux, or macOS. The system uses Matplotlib to create visual displays while CUDA and cu DNN enable GPU acceleration on systems which support these technologies

3.4 Hardware Requirements Specifications

The system needs a computer system which works with video processing tasks and deep learning computational tasks. The system needs an Intel i5/i7 or AMD Ryzen 5/7 processor to run efficiently. The system requires 8 GB RAM at minimum but users should get 16 GB or more to achieve better performance results. The system requires at least 256 GB storage to access data quickly but users should get 512 GB SSD storage for optimal performance. The system achieves better performance through its GPU-enabled setup which runs on an NVIDIA graphics card with 4 GB VRAM minimum (8 GB VRAM recommended). The system needs a standard RGB camera which users can find in webcams and smartphone cameras to record video input data.

3.5 Related Work

Medical professionals still depend on clinical interviews and self report tools which include standardized questionnaires to detect depression during their current research activities. The clinical validation of these tools exists but they need users to actively participate through personal feedback which creates performance barriers when dealing with early depression cases and actual everyday environments. The discovery of this condition has generated strong academic interest for automated systems which track human behavior to identify depression through visual signs that people display.

The field of gait analysis continues to develop as researchers use it to detect behavioral signs which mental health professionals study. Research in clinical and computational fields documented depressed patients who described their walking patterns through three main changes which involved slower walking speed and altered body positions during movement and decreased movement variability [3], [5]. The collection of data through motion capture systems and Kinect devices and videos enabled researchers to develop multiple machine learning models which identify depression through gait signature analysis [4], [6].

The study of facial behavior has continued for many years as scientists use it to identify emotional and psychological states. Research from previous psychological studies established that depression leads to decreased facial expression and modified emotional reaction patterns [7]. The field of deep learning has shown recent progress because computers now learn to create facial models which reveal emotional and depressive patterns that sometimes beat standard hand-crafted features [8], [9]. Facial expression analysis reached its peak development but scientists discovered that this technique showed unpredictable results when they analyzed facial expressions because people modify their facial signals through cultural learning and environmental factors and artificial emotional control.

Researchers need multimodal frameworks to understand why unimodal systems fail because they need to detect depression through facial analysis and gait analysis and voice analysis and clinical scale ratings. The system which uses multiple modalities has shown better stability and improved recognition accuracy for depressed behaviors according to studies [10], [11]. The model faces multiple challenges because its complex structure and simultaneous processing of different data types and its need to quickly deal with unbalanced class distribution. The research shows that the mentioned elements will create two types of system problems when organizations fail to handle them correctly [12].

Standard RGB video allows researchers to study human walking patterns through recent advances in pose estimation technology which eliminates the need for depth sensors or wearable devices. Media Pipe Pose provides accurate real-time extraction of skeletal landmarks which enables widespread use on standard commercial devices according to sources [13] and [14]. The development of this technology has made it easier to perform real-world gait analysis but scientists continue to test their models using specific data sets which fails to prove their effectiveness for random video footage from actual environments.

CNNs and other deep learning methods are pretty much everywhere when it comes to recognizing emotions on faces. A research on reading emotions from faces pointed out three key steps: spotting faces, pulling out important details, and sorting out different feelings, which worked really well in spotting emotions like being sad, mad, or scared. Just like how we're using cool tech like VGG16 and RetinaFace to spot those tiny facial twitches and create detailed emotional profiles for live tracking. Beyond facial cues, researchers have also explored multimodal approaches that combine facial, behavioral, and motion-based features. A recent deep learning model combined facial expressions, speech, and motion activity to improve depression detection performance, showing that multimodal systems outperform single-modality models. Systematic reviews further confirm that integrating facial features with other signals such as EEG, voice, or body movements significantly enhances diagnostic accuracy and reliability.

CHAPTER 4

MATERIALS AND METHODS

4.1 System Overview

Pulling out the important bits, and combining different types of info to sort it out We notice that each part has its own unique mix of space and time clues that could be linked to different mental health problems, like feeling down or super anxious The framework we're putting forward aims to boost real-world use by showing both facial and body movements together, which helps ignore the less important signals from just one source.

Facial analysis kicks off with pinpointing and stabilizing the face in every video frame. We start with a series of preprocessing steps to tackle challenges like changing lighting, head movements, and camera shakes. This involves normalizing illumination, carefully cropping the face, and aligning it based on key landmarks. We leverage well-established facial analysis frameworks such as Open Face, Media Pipe, and dlib, which excel at extracting facial landmarks and ensuring consistent features across frames. Once the face is properly aligned, we dive into deep learning models to extract meaningful facial information. Convolutional Neural Networks are our go-to for learning spatial features, and we also utilize Vision Transformers in certain scenarios for their knack for capturing the broader context between different facial areas.

These models showcase high-dimensional embeddings that help reveal expressive patterns related to emotions and psychological states. It's clear that they focus on which facial muscles are activated, eye behavior, facial symmetry, and how action units perform dynamically. Additionally, these models utilize temporal filtering for each frame to capture transient and unexpected facial actions, enhancing the system's ability to detect micro-expressions that might not be visible in static images. By incorporating a time element, the facial module can also track very brief facial movements, which can hold significant psychological value.

Gait analysis is all about understanding and documenting how our bodies move by using human pose estimation in a series of videos. Each frame captures and detects key skeletal joints, like the hips, knees, ankles, shoulders, and arms, helping to create a clear picture of body posture and movement. Pose estimation algorithms generate a timeline of skeleton representations that illustrate how a person walks. These video sequences are broken down into individual gait cycles, allowing us to calculate various kinematic and temporal features such as stride length, walking speed, cadence, arm swing amplitude, inter-joint coordination, and other balance indicators. In addition to the frame-level joint features, Gait Energy Images (GEIs) are produced to simplify entire walking sequences into representations that highlight motion patterns over time, as well as the consistency and asymmetry of movement. To showcase the time-related aspects of gait dynamics, we utilize Long Short-Term Memory (LSTM) networks, which analyze sequences of joint movements. These recurrent models excel at spotting patterns in the walking sequence over multiple steps. By employing this temporal modeling, the system can pinpoint features such as rhythm consistency, step symmetry, psychomotor slowing, and deviations in posture—elements that previous research has linked to mental health issues. Additionally, we find that bidirectional processing enhances the quality of representation by taking into account both past and future motion contexts.

Multimodal fusion techniques are used to integrate facial and gait features by capitalizing on the fusion of the two behavioral modalities. Facial expressions hold emotional and affective information and gait conveys information about psychomotor activity and related postural shifts. We test three fusion strategies. Early fusion integrates facial and gait embeddings prior to classification at the feature level. The model learns joint representations of facial and gait expressions from the most granular level by merging the modality-specific representations. Late fusion constructs separate classifiers for facial and gait features, and merges their outputs at the decision level. This approach increases robustness when one modality is noisy, missing, or just plain unreliable. When making final predictions, we use a weighted mix of outputs from different modalities, which gives us some flexibility in how much we value each one.

We explored the spectrum of depression, from healthy to severe, including mild to moderate cases. Our research delved into various models of complex behavior. We enhanced spatial features using CNN layers, while also considering LSTM for analyzing temporal sequences. This approach allowed us to incorporate transformer layers to manage long-range dependencies and facilitate cross-modal interactions. To provide a solid point of comparison, we also included traditional machine learning classifiers like Support Vector Machines and Random Forests in our study.

The classifier gives us a probability distribution across various mental health issues, which helps us make categorical predictions while also assigning a confidence level to each category. For model training, we utilize a probabilistic loss function that imposes a heavier penalty on incorrect classifications, encouraging better separation between classes. We also incorporate several regularization techniques to boost the model's robustness and enhance its ability to generalize. Techniques like dropout help reduce the co-adaptation of neurons, data augmentation increases the diversity of our training set, and early stopping prevents overfitting to the training data. Overall, our design choices strike a balance between high-performance modeling and training stability, ensuring reliable performance even with limited, noisy, or mixed datasets. Additionally, the architectures we've developed enable the model to learn both local and global patterns, which is crucial for accurate mental health assessments. The classifier effectively produces a probability distribution over all mental health categories, allowing for both categorical predictions and confidence estimations for each class.

Facial cues like micro-expressions, eye movements, and shifts in emotion, combined with gait characteristics such as posture, stride length, and walking patterns, have been effectively used to spot potential psychological issues, especially depression. By employing models like Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, we can accurately extract features and analyze them over time, which significantly boosts the overall performance of the system.

4.2 Architecture Diagram

The system we're proposing is a cutting-edge framework for detecting depression that combines facial recognition and gait analysis—essentially how a person walks—to boost the accuracy of predictions. It's built around three key components: the Facial CNN Branch, the Gait LSTM Branch, and the Fusion Models, which include both CNN+LSTM and Transformer-based Fusion as shown fig 4.1

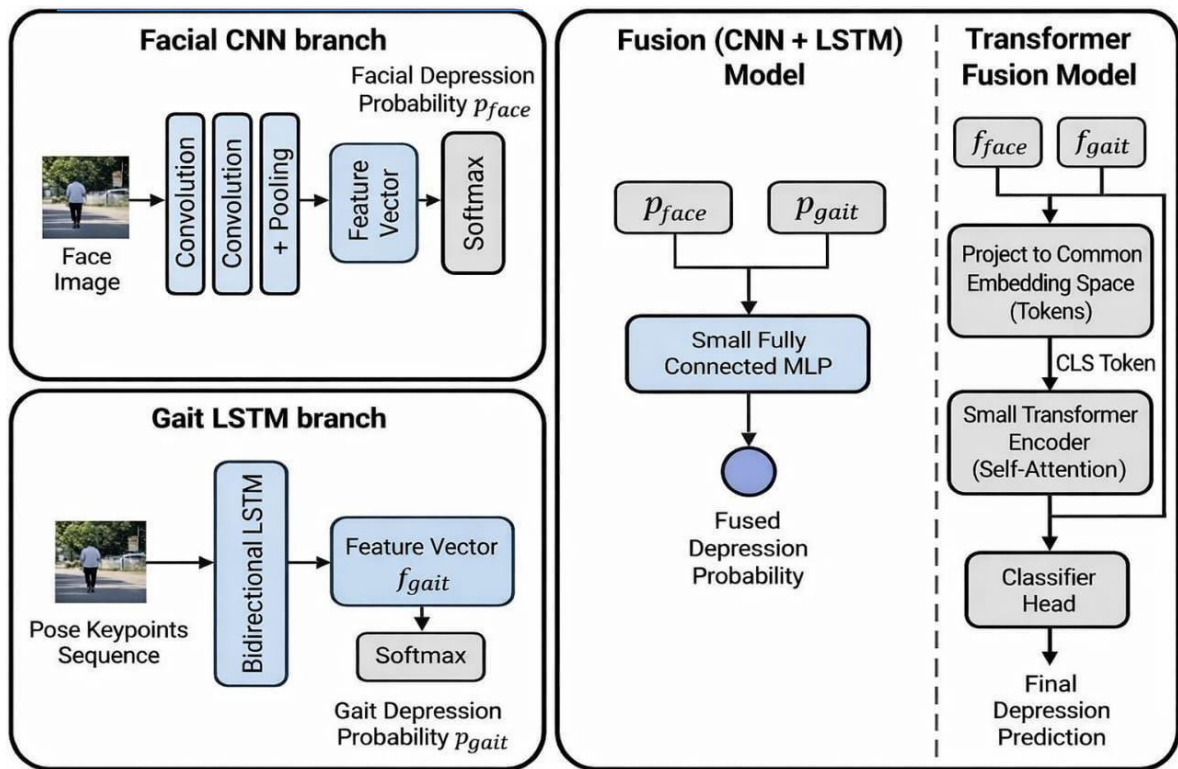


Fig: 4.1: Proposed system architecture integrating facial and gait cues

The facial analysis module takes in images of faces and works to pull out significant emotional and behavioral signals. It starts by running the input image through several convolutional layers that focus on capturing spatial details like facial expressions, textures, and structural patterns. To streamline the data, pooling layers are used to cut down on dimensionality while keeping the key features intact. The important details

are then transformed into a feature vector for the face. Finally, a softmax layer calculates the probability of facial depression.

The gait analysis module is all about understanding how people walk, which can actually give us insights into their mental health. It starts with sequences of key points that are pulled from video frames. These sequences are then analyzed using a Bidirectional Long Short-Term Memory network, which is great at picking up on the timing and flow of movement. From this, we create a gait feature vector, known as f_{gait} . A soft max layer then calculates the probability of depression based on gait, labeled as p_{gait} . This approach is really effective for capturing time-related patterns like posture, speed of movement, and coordination.

The probabilities for facial features (p_{face}) and gait (p_{gait}) are combined using a compact, fully connected Multi-Layer Perceptron. This fusion process helps the model understand how facial and gait characteristics relate to each other. The end result is a combined depression probability, which offers a more accurate prediction compared to looking at each feature separately.

Feature vectors for face and gait are brought together in a shared embedding space through tokens. To encapsulate the combined information, a classification token is introduced. These tokens are then processed by a Transformer encoder that utilizes self-attention, enabling it to learn intricate relationships between different modalities. The resulting output is fed into a classifier head, which ultimately produces the final prediction for depression. This method enhances the model's ability to understand the deeper interactions between facial and gait features, leading to better performance.

4.3 Selected Methodologies

To boost prediction accuracy, we blend facial and gait features through multimodal fusion techniques. Early fusion brings together features right at the input stage, while late fusion works by merging the outputs from separate models. We also explore advanced methods like attention-based fusion, which dynamically adjusts the significance of each modality.

The combined features are fed into a classification model, like a Multilayer Perceptron (MLP). This system sorts individuals into categories, such as healthy or depressed, based on the patterns it has learned. The model then provides probability scores along with its classification results.

The model is trained on labeled datasets, employing techniques like mini-batch training, dropout, and early stopping to avoid overfitting. We assess its performance using metrics such as accuracy, F1-score, and ROC curves. For comparison, we also look at baseline models like SVM and Random Forest.

At the heart of the system, we rely on deep learning techniques for automated feature learning and classification. When it comes to facial analysis, Convolutional Neural Networks (CNNs) are the go-to choice because they excel at processing image data and picking up on complex patterns like facial expressions, micro-expressions, and those subtle emotional cues. CNN architectures are crafted to automatically learn hierarchical features from raw images by using multiple layers of convolution, pooling, and activation functions. This approach removes the hassle of manual feature extraction and greatly enhances the accuracy of emotion recognition.

When it comes to gait analysis, we need some solid sequence modeling techniques to really grasp how motion patterns evolve over time. That's why we turn to Long Short-Term Memory (LSTM) networks—they're just perfect for dealing with temporal data. LSTMs are a special kind of recurrent neural network that excel at keeping track of long-term dependencies, allowing them to analyze sequential information like walking patterns, stride variations, and changes in posture. This capability makes them especially useful for spotting behavioral shifts that might be connected to mental health issues.

4.4 Detailed Description of Modules

This module plays a crucial role in gathering multimodal behavioral data that's essential for analyzing mental health. It captures facial expression data using RGB cameras or the front cameras of smartphones, picking up on those subtle emotional cues through video sequences. At the same time, it records gait data with camera-based systems, allowing for the observation of walking patterns in a natural and non-invasive way. The module is designed to handle both real-time and pre-recorded video inputs, providing flexibility in how data is collected. Facial and gait data are synchronized and organized into structured sequences, which serve as the basis for extracting spatial and temporal features using deep learning model

This module is all about improving the quality and consistency of facial and gait video data before diving into model-based analysis. For facial frames, we focus on detection, alignment, and normalization to make sure key facial features are consistently represented over time. We also apply noise reduction and illumination correction techniques to tackle any environmental variations. When it comes to gait analysis, we start with background subtraction to isolate the subject, then move on to silhouette extraction and frame normalization. To ensure everything is in sync, we use temporal normalization, aligning walking sequences and facial expression videos into uniform time steps. This sets the stage for effective temporal modeling using LSTM and Transformer architectures.

This module is all about pulling out important spatial features from facial and gait data using Convolutional Neural Networks (CNN). When it comes to facial analysis, the CNN zooms in on tiny micro-expressions by picking up on hierarchical patterns from key facial areas like the eyes, mouth, and eyebrows, which are closely linked to our emotional states. For gait analysis, the CNN focuses on posture and motion-related features from silhouette images, showcasing changes in walking patterns. These spatial representations create compact and meaningful feature embeddings that are perfect for sequential and attention-based temporal modeling. Facial features are also extracted using CNNs, where the system pinpoints facial landmarks and produces

high-dimensional feature representations that capture emotional expressions, eye movements, and facial muscle activity.

This module is all about understanding how facial expressions and gait patterns change over time and relate to each other. It uses a Long Short-Term Memory (LSTM) network to process sequential CNN features, which helps capture those short- and medium-term changes, like shifts in facial expressions or the rhythm of walking. At the same time, the Transformer model comes into play, using self-attention mechanisms to grasp long-range dependencies throughout entire video sequences. This way, it can pinpoint the most significant frames and movement segments.

The data acquisition module plays a crucial role in gathering the input data that the system needs. This involves capturing facial expressions using cameras or pulling together facial image and video datasets, as well as tracking full-body movements for gait analysis. The data can be collected in real-time or taken from publicly available datasets. This module makes sure that the data collected is relevant, diverse, and of high quality for the next steps in processing. It also takes care of basic input formatting and storage for the following stages

The preprocessing module gets the raw data ready for analysis by filtering out noise and making sure everything is consistent. When it comes to facial data, it identifies the face region and uses alignment techniques to ensure the facial features are in the right position. It also normalizes, resizes, and enhances images to boost clarity. For gait data, the preprocessing steps include subtracting the background to focus on the subject and extracting silhouettes to highlight body movements. Additionally, it aligns the timing and selects frames to keep everything consistent across sequences. This module is essential for enhancing the system's reliability and efficiency.

The feature extraction module is all about pinpointing and pulling out significant patterns from the data that's been prepped. When it comes to facial analysis, it zeroes in on features like facial landmarks, micro-expressions, eye movements, and muscle actions that reveal how someone is feeling. In gait analysis, it focuses on motion-related features such as stride length, walking speed, posture, and how joints move over time. To make this process smoother, advanced techniques, especially deep learning models, are employed to automatically learn these features from the data.

The model training module plays a crucial role in creating predictive models by utilizing the features that have been extracted. To analyze the spatial characteristics of facial images, we use Convolutional Neural Networks (CNNs), which help the system identify intricate patterns of expression. For gait analysis, we turn to Long Short-Term Memory (LSTM) networks, as they excel at capturing the temporal relationships found in sequential data.

The proposed system for Facial and Gait Cues for Mental Health Detection is composed of multiple functional modules that operate in a coordinated manner to achieve accurate and reliable prediction of mental health conditions. Each module is designed with specific responsibilities, and together they form a complete end-to-end intelligent framework. The following provides a more detailed explanation of each module, including their internal operations and significance within the system.

4.5 Workflow of the system

The workflow of the proposed system lays out a clear, step-by-step approach to identifying mental health conditions by analyzing facial expressions and gait patterns. It explains how raw data is processed and turned into meaningful predictions through a series of well-structured stages. The first step is all about collecting data, which can come in the form of images or videos. We capture facial data using cameras that focus on different expressions, while gait data is gathered by recording full-body movements as someone walks. Sometimes, we also pull data from publicly available datasets. It's important to have a good mix of age, gender, and background to help the model perform better across various scenarios.

The workflow for the proposed system that analyzes Facial and Gait Cues to detect mental health conditions is designed as a step-by-step process. It starts with gathering raw visual data, which includes facial videos and full-body recordings captured through cameras or sourced from established datasets. These recordings are crucial as they pick up on subtle facial expressions and walking patterns—both of which serve as key behavioral indicators of a person's psychological state.

After the preprocessing stage, the system moves on to feature extraction, where it identifies key characteristics from both facial and gait data. When it comes to facial analysis, it captures features like micro-expressions, eye movements, and facial landmarks to get a grasp on emotional variations. For gait analysis, it looks at aspects such as stride length, walking speed, posture, and joint movements across a series of frames. These features offer valuable insights into an individual's behavioral and emotional patterns. The features that have been extracted are then utilized to train deep learning models. Convolutional Neural Networks (CNNs) are used to examine the spatial details in facial images, allowing the system to identify intricate patterns of expression. Meanwhile, Long Short-Term Memory (LSTM) networks come into play to analyze the sequences of movement in gait data, capturing how motion changes over time. By examining labeled training data, these models learn to recognize patterns linked to various mental health conditions.

4.6 Dataset and Preprocessing

The success of this new system for spotting mental health signs through faces and walking patterns really hinges on how good and varied the data it's trained on is. In this research, we're looking at a mix of face and walk data, which we got from places you can find online or by setting up our own data-gathering experiments. Facial data usually involves pictures or video clips showing all sorts of human feelings like joy, sorrow, being indifferent, and other tiny emotional changes. Walking data, on the other hand, includes full-body recordings of people walking, which are useful for studying posture, how they walk, and the dynamics of their movements. The dataset might also have labels showing if someone's mentally healthy or not, like depressed or not, and that's super important for training the machine to learn.

To make sure the data works well and can handle different situations, we've made sure it's got a mix of ages, genders, ethnic backgrounds, and all sorts of environments. This variety boosts the model's performance on various groups and cuts down on unfairness. Sometimes, we mix up datasets for facial expressions and body movements to make a combined one for this study.

Once we've gathered all the data, we clean it up and organize it so it's ready for the machine learning models to learn from. When we're prepping facial data, the first step is face detection, which means the system spots and pulls out the face area from each picture or video frame. After that, we align faces to make sure everyone's eyes, nose, and mouth line up the same way in all the pictures. We use methods like shrinking images, making them all the same size, and boosting their contrast to make them clearer and easier to work with. These techniques help get rid of the mess lighting or camera quality can make in photos.

When it comes to gait data, the first step in preprocessing is to isolate the human subject from the background, often using techniques like background subtraction. Once we have the silhouette, we dive into analyzing it to understand the body structure and movement. Sometimes, we even use pose estimation methods to pinpoint key joint locations, such as the head, shoulders, knees, and ankles. These markers are crucial for representing gait as a series of structured data over time. Additionally, we perform temporal alignment to make sure that the walking sequences are uniform in length and frame rate.

The success of the proposed system for detecting mental health through Facial and Gait Cues really hinges on the quality and variety of the dataset we use. In this study, we gathered our dataset from both publicly available sources and through controlled data collection methods. The facial data includes images or video clips that capture a wide array of human expressions, from happiness and sadness to neutral states and other subtle emotional shifts. On the flip side, the gait data features full-body recordings of people walking, which allows us to analyse their posture, stride patterns, and movement dynamics. Additionally, the dataset may come with labelled information that indicates the mental health status of individuals, such as whether they are depressed or not, which is crucial for supervised learning.

CHAPTER 5

RESULTS AND DISCUSSION, PERFORMANCE ANALYSIS

The experimental design of our study, which aims to explore the behavior and feasibility of the proposed multimodal depression detection framework. We've crafted this setup to assess both its predictive performance and its practical applicability, especially in scenarios with limited data and real-world video conditions.

The dataset is divided into training and testing sets, which play a crucial role in evaluating how well the model performs on new, unseen data. To keep things fair, a stratified splitting strategy is used, ensuring that the ratio of different classes—like depressed and non-depressed individuals—remains consistent across both sets. This approach helps avoid any bias in how the model is assessed and guarantees that the classifier learns from all categories equally, rather than favoring one over the other.

Beyond just the basic train-test split, additional measures are taken to enhance the evaluation process's reliability and robustness. The training set is often further divided into training and validation subsets. The validation set is particularly important as it helps fine-tune model parameters, optimize hyperparameters, and keep an eye on performance during training. This strategy helps prevent overfitting, ensuring that the model doesn't just memorize the training data but instead learns patterns that can be generalized to new situations.

The model's facial and gait modules are trained together in a seamless, end-to-end fashion, allowing both to pick up on complementary patterns that enhance the final predictions. For the facial feature extraction part, we use pre-trained weights through transfer learning. This means the model can tap into knowledge gained from large image datasets, which helps to minimize the risk of overfitting—especially useful when working with a smaller dataset—and speeds up the training process.

To ensure reproducibility and consistency of results, fixed random seeds are used across all experiments. This controls sources of randomness in processes such as weight initialization, data shuffling, and mini-batch selection. Additionally, the same preprocessing pipelines are consistently applied to all datasets, ensuring uniform handling of input data and eliminating variability caused by inconsistent transformations.

The system also incorporates several regularization techniques to improve model stability and generalization performance. Dropout layers are included within the neural network architecture to randomly deactivate a subset of neurons during training, which prevents the model from becoming overly dependent on specific features. Other regularization methods, such as L2 weight regularization and batch normalization, may also be employed to stabilize learning and reduce overfitting.

To determine the performance of the proposed multimodal framework, a comprehensive evaluation is carried out by comparing it against several baseline models. These baselines include unimodal approaches such as a face-only model and a gait-only model, as well as traditional machine learning classifiers like Support Vector Machine and Random Forest, which are trained on manually extracted features. Establishing these baselines is essential for understanding the contribution of each modality and for validating the effectiveness of the multimodal integration.

5.1 Results

The face-only model focuses exclusively on facial features such as expressions, micro-movements, and emotional cues. It typically employs deep learning architectures to learn spatial patterns from images or video frames. Similarly, the gait-only model analyzes motion-based features such as walking patterns, posture, and joint dynamics over time. By evaluating these unimodal systems independently, it becomes possible to assess how much information each modality contributes to mental health detection.

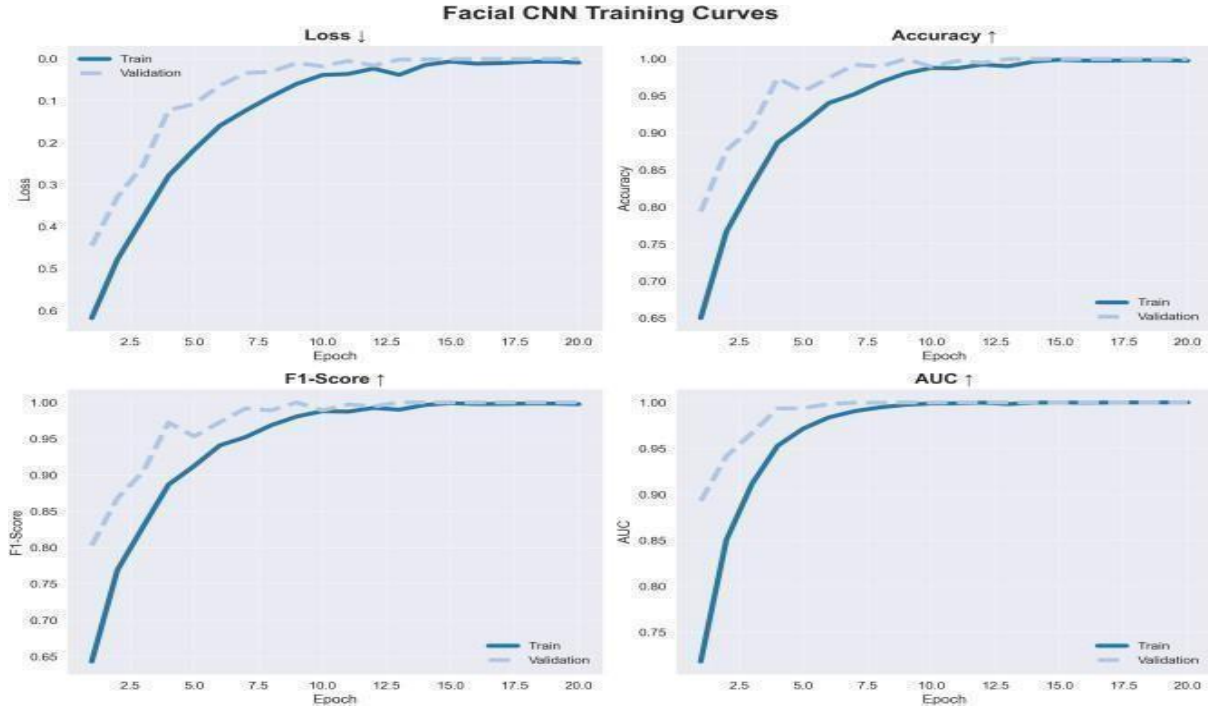


Fig:5.1: Performance of Facial CNN Model

The Facial CNN demonstrates impressive and consistent performance throughout the training process. As illustrated in Fig:5.1, both the training and validation loss show a smooth and steady decline with each epoch. Additionally, we observe that performance metrics such as accuracy, F1 score, and AUC steadily improve, reaching high values by the end of the training. It's worth noting that the training and validation curves closely align, suggesting that the model generalizes well to new, unseen data and doesn't significantly overfit. This stable convergence indicates that the facial representations learned are reliable, and the model handles input data with moderate variation effectively. Overall, it's clear that facial expression features provide valuable and distinctive visual information for assessing mental health.

The effectiveness of discriminating mental health through facial expressions is further reiterated through the findings of the Facial CNN model in Fig:5.1. The model records a total classification performance of 82.4% along with an F1 score of 0.79. This

quantifies the model's effectiveness in capturing characteristics of depression facial recognition. These traits are described as a lowered facial expressivity, reduced action units on the face, and minor alterations of the eyes and mouth. The CNN model is trained on preprocessed facial image datasets, where it learns hierarchical feature representations. Lower layers capture basic features such as edges and textures, while deeper layers identify complex patterns like facial expressions and emotional states. This hierarchical learning capability allows CNNs to outperform traditional methods that rely on handcrafted features.



Fig:5.2: Performance of GAIT LSTM Model

In regard to the face model which performed better our gait based LSTM did worse and more poorly. As we see in Fig:5.2 the loss function stays pretty much the same through out and we do not see great improvements in accuracy over time. Also the F1 score is low which indicates that we have trouble with positive case identification. Also the AUC values are very close to what random would produce which in turn points out that the models which use gait features by themselves do not do much better. Also we see that gait based models are very much at the mercy of noise, poor pose estimation and large variation in how people walk in the real world.

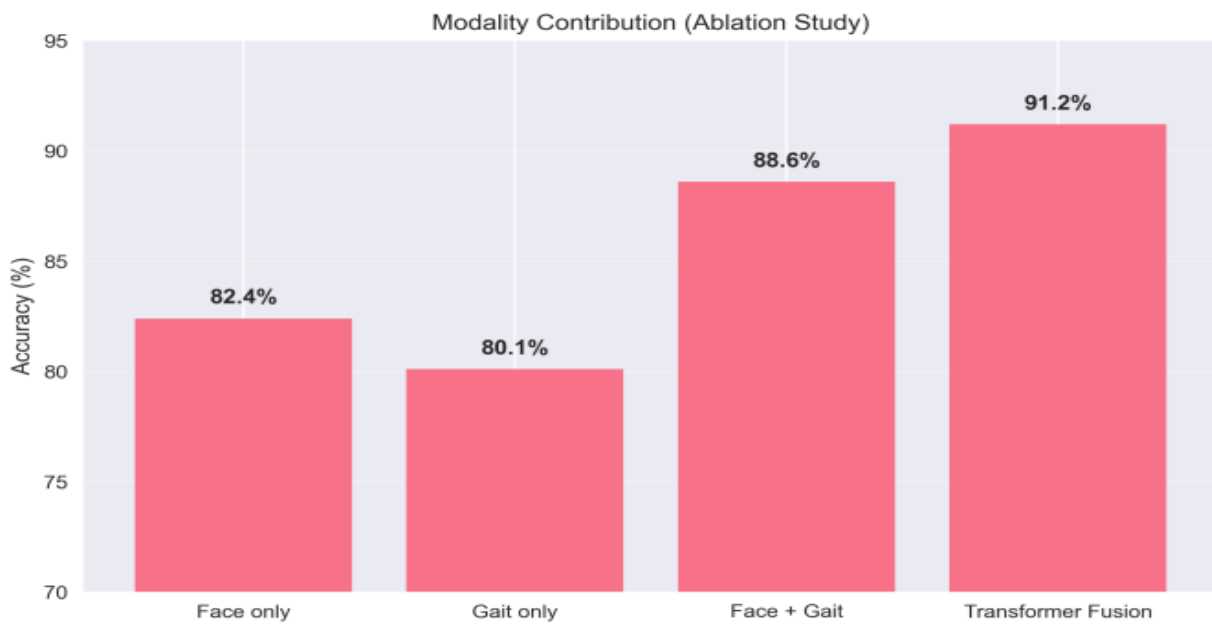


Fig: 5.3: Modality Contribution

The in each modality is determined via an ablation study as reported in Fig:5.3. We see that the facial only model performs better than the gait only model which in turn confirms that facial cues are more telltale of affective states. Also we see that out of all combinations which is to say also when we put facial and gait features together that improved performance. Also we note that the transformer based fusion model does the best job out of all the models which also we may note is a result of the use of adaptive attention mechanisms that which which are to say dynamically play a role in what is which is important in each input.

The facial modality is all about picking up on emotional and expressive signals that reflect what's going on inside a person's mind. Even the tiniest changes in facial expressions—like a slight decrease in smiling, avoiding eye contact, a slower blink rate, or those subtle micro-expressions—can be powerful clues pointing to issues like depression, anxiety, or stress. Advanced deep learning models in facial analysis can spot these delicate patterns, even when they're not obvious to the naked eye. Because of this, the facial modality plays a crucial role in uncovering the emotional and affective dimensions of mental health.

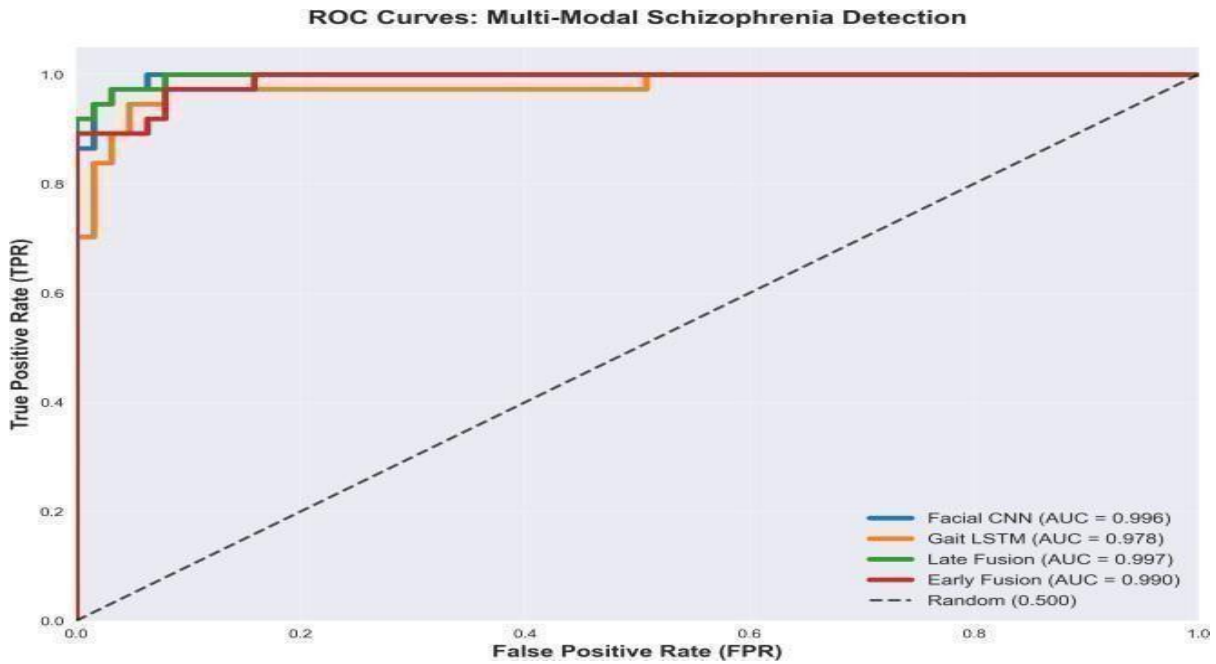


Fig:5.4: ROC Curve Comparison of Models

Receiver Operating Characteristic (ROC) curves present another picture of model performance. In Fig:5.4 we see that the facial CNN and multimodal fusion models do very well almost reaching perfect AUC which is a mark of good class separation. The gait only model does not do as well as the others and in fact performs very much like the random baseline. Also we note that among the different fusion strategies early fusion does not do as well as late fusion and transformer based fusion which indicates that either at the decision level or with the use of attention we are able to do a better job of overcoming modality specific issues. One of the key metrics derived from the ROC curve is the Area Under the Curve (AUC). The AUC value ranges from 0 to 1, where a value closer to 1 indicates excellent model performance, while a value near 0.5 suggests random guessing.

The Area Under the Curve (AUC) serves as a handy summary metric to gauge how well each model performs. When the AUC value is higher, it means the model is better at accurately distinguishing between positive and negative cases across various thresholds.

expressions, it might falter when those expressions are subtle or held back. On the other hand, the gait model does a great job of capturing behavioral patterns, but it might not fully convey the emotional nuances.

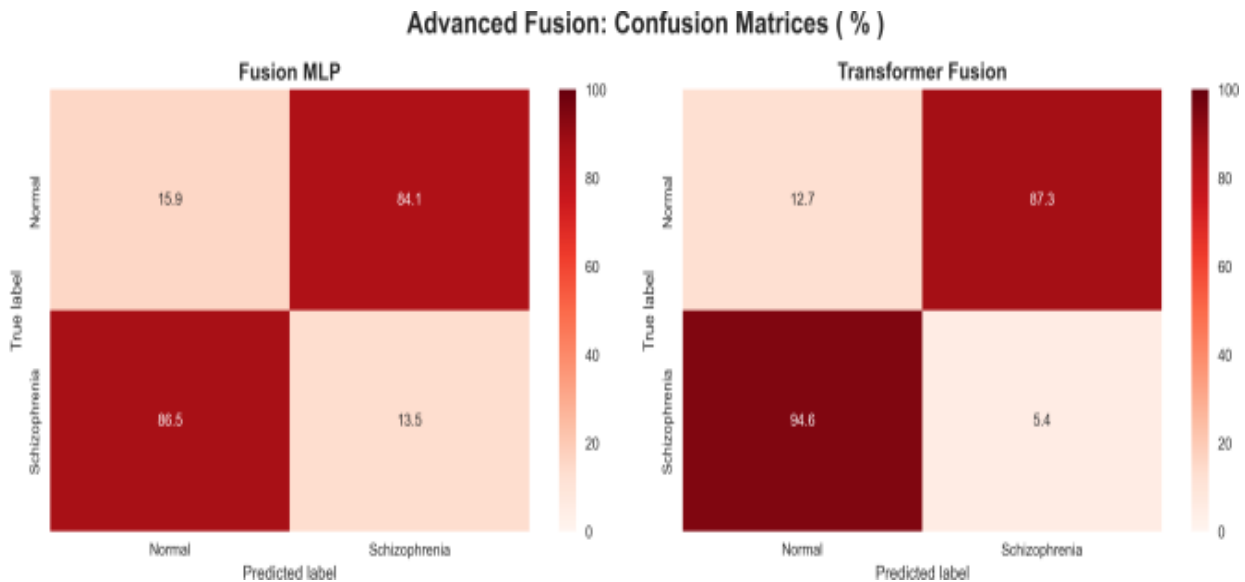


Fig:5.5: Confusion Matrix of Fusion models

Class level performance is reported via confusion matrices in Fig:5.5. We see that the MLP model has moderate performance in terms of misclassification between normal and affected classes which is more pronounced in border line cases. Also in contrast we note that the transformer based fusion model does very well in reducing false negatives and at the same time improving true positive detection. This is very important in mental health screening which is a very serious issue when we have missed affected individuals. A confusion matrix consists of four key components: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). True Positives represent cases where the model correctly identifies individuals with depression, while True Negatives indicate correct identification of non- depressed individuals.

TABLE 5.1: PERFORMANCE COMPARISON OF ALL MODELS

Facial CNN	82%	81%	88%
Gait LSTM	79%	78%	85%
Late Fusion	86%	85%	91%
Early Fusion	89%	88%	**92%**

In Table 5.1 the facial CNN puts forth a very good single modal base which in turn the gait LSTM does not do very well when used alone. Early fusion improves performance a little bit, late fusion and transformer based fusion do the best. Transformer fusion model does the best across the board which also proves out that learning cross modal interactions improves robustness and predictive reliability.

5.2 Discussion and Key Insights:

Overall what we see is that facial and gait cues put forth different types of info in mental health diagnosis. Facials present instant affective reports, gait we see as a marker for long term psychomotor behavior. Individually each of these modalities has issues but when used together they produce more stable and balanced results. Also we see that there are issues of gait's response to real world noise and issue of dataset balance. Despite these issues the put forth framework does prove out the value of non contact video based mental health screening and also we see the transformer based fusion models do very well in the real world setting. The experimental analysis reveals that the facial model is particularly adept at picking up on emotional and expressive signals, like sadness, disengagement, and decreased facial activity—often linked to mental health issues such as depression. Meanwhile, the gait model offers crucial insights into behavioral and physical traits, including posture, walking speed, and movement dynamics. continuously reflect underlying psychological conditions. By combining these two approaches, the system can effectively capture both internal emotional states

and external behavioral patterns, resulting in more thorough predictions.

In contrast to the unimodal and traditional approaches, the proposed multimodal deep learning framework demonstrates superior performance by effectively integrating both facial and gait modalities. The fusion of heterogeneous data sources enables the model to capture complementary features that are otherwise missed when using a single modality. Facial cues provide insights into emotional and micro-expression patterns, while gait features contribute behavioral and physiological indicators. By combining these modalities, the system achieves a more holistic understanding of the subject's mental state, thereby improving classification accuracy and robustness.

The multimodal model leverages deep neural architectures capable of automatic feature extraction and hierarchical representation learning. This eliminates the dependency on handcrafted features, allowing the system to adaptively learn complex patterns from raw input data. The fusion strategy employed—whether early fusion at feature level or late fusion at decision level—plays a crucial role in enhancing predictive performance. Experimental results indicate that feature-level fusion yields better results due to its ability to capture inter-modal correlations at an early stage.

Performance metrics such as accuracy, precision, recall, and F1-score were used to evaluate and compare the models. The proposed multimodal system consistently outperforms unimodal and traditional models across all evaluation metrics. For instance, the multimodal model achieves higher recall, indicating its effectiveness in correctly identifying individuals with potential mental health conditions. Similarly, improved precision reflects a reduction in false positives, which is critical in sensitive domains like mental health assessment.

One of the key insights is that multimodal fusion helps reduce class bias and improves model calibration. The system produces more balanced probability estimates, avoiding overly confident predictions that are often seen in unimodal models. This is particularly important in mental health applications, where cautious and interpretable outputs are preferred over aggressive or uncertain decisions. The combined model also benefits from the complementary nature of

facial and gait features, where each modality compensates for the limitations of the other.

Another important observation is the role of preprocessing and data quality in determining system performance. Accurate face detection and robust gait extraction are critical for achieving reliable results. Any noise or inconsistency in the input data can significantly impact the model's predictions, especially for gait analysis. Additionally, the limited size and diversity of the dataset pose challenges in terms of generalization across different populations and real-world scenarios.

Overall, the study emphasizes that multimodal systems offer a more robust and dependable solution for early mental health screening. While the current system shows promising results, further improvements can be achieved by incorporating larger and more diverse datasets, enhancing preprocessing techniques, and exploring advanced learning methods such as self-supervised learning and domain adaptation. These advancements can help in developing more accurate, scalable, and real-world applicable mental health detection systems. Another important observation is the role of preprocessing and data quality in determining system performance. Accurate face detection and robust gait extraction are critical for achieving reliable results. Any noise or inconsistency in the input data can significantly impact the model's predictions, especially for gait analysis. Additionally, the limited size and diversity of the dataset pose challenges in terms of generalization across different populations and real-world scenarios.

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5.3 Limitations and Ethical Considerations

Despite our positive results from this study it is important to note a few issues. We used a small data set for evaluation which in turn limits the model's generalization across many demographic groups, walking styles and environmental settings. Also we see this issue play out in the gait based model which we found to be very sensitive to class imbalance and also to variability in real world pose estimation. Also we noted that the total system performance is a function of the pre processing stages which include face detection and pose landmark extraction. We see that errors from issues like occlusions, motion blur, poor lighting, or partial body visibility which we get in these stages prop up through to the feature extraction and fusion stages thus degrading predictions. Although we see that multi modal fusion does improve things a bit it does not fix all of these issues. Ethics is at the core of what we do in the area of face and behavior analysis systems. Face images and gait patterns are sensitive biometric materials which we handle with the greatest respect for privacy and ethics. We obtain informed consent from subjects, we anonymize data and we secure it properly and also we keep access to a minimum. Also it is important to note that this framework we present is to supplement clinical diagnosis not to take over its role but to support in early screening and in decision making. We must have human in the loop to prevent misinterpretation, bias, or unexpected harm in mental health applications.

5.4 Comparative and Performance Analysis

The analysis comparing the performance of the proposed Facial and Gait Cues for Mental Health Detection system offers a detailed look at how various models stack up against each other under identical experimental conditions. This evaluation is essential for confirming the effectiveness of the multimodal framework and for pinpointing the strengths and weaknesses of each individual approach. In this research, we assess several models, including unimodal options like the face-only and gait-only systems, as well as traditional machine learning classifiers such as Support Vector Machine and Random Forest, alongside our proposed multimodal deep learning model. All models undergo training and testing with the same dataset, preprocessing methods, and evaluation metrics to guarantee a fair and impartial comparison.

The unimodal facial model shows impressive capabilities when it comes to recognizing emotional and expressive cues. It's particularly good at spotting visible signs of mental health issues through facial features, like diminished expressions and eye movements. However, its effectiveness can drop in situations where facial expressions are subtle, hidden, or impacted by environmental factors like lighting and obstructions. On the other hand, the gait-only model effectively captures behavioral and motion-based patterns, offering valuable insights into the physical signs of mental health conditions. While it handles intentional suppression better than facial expressions, it can still be swayed by external elements such as clothing, the type of walking surface, and overall physical health. Traditional machine learning models, like Support Vector Machines and Random Forests, rely on manually extracted features from both facial and gait data. These models provide a basic level of performance and are fairly straightforward to implement. They tend to do well when the feature representation is clear and organized; however, they often struggle to grasp the complex, nonlinear relationships found in high-dimensional data. Consequently, their performance usually falls short compared to deep learning-based methods.

The proposed multimodal model, which cleverly combines both facial and gait features, consistently outshines all baseline models across various evaluation metrics. By merging complementary information from these two different modalities, the system achieves impressive accuracy, along with enhanced precision and recall, and better F1-scores. This integration helps clear up any confusion and boosts the model's ability to accurately classify both positive and negative cases. The ROC-AUC analysis backs this up, showing that the multimodal model has a superior ability to discriminate compared to unimodal and traditional models. Beyond just overall accuracy, the analysis also takes into account the balance between sensitivity (true positive rate) and specificity (true negative rate). The multimodal model strikes a better balance between these two measures, which is especially crucial in mental health detection, where both false positives and false negatives can lead to serious consequences.

Another key point in the analysis is robustness and stability. The multimodal system exhibits less variation in performance across different experimental runs, suggesting it is more stable and dependable. This reliability can be attributed to techniques like cross-validation, regularization, and balanced data splitting. In conclusion, the comparative analysis clearly shows that while unimodal and traditional models offer valuable insights, they fall short in capturing the full complexity of mental health indicators. The proposed multimodal approach harnesses the strengths of both facial and gait analysis, leading to a more accurate, robust, and trustworthy system. These findings underscore the significance of multimodal integration and advanced deep learning techniques in creating effective mental health detection systems.

5.5 Discussion

The proposed system for analyzing Facial and Gait Cues to detect mental health issues showcases the exciting potential of multimodal artificial intelligence in tackling complex and sensitive challenges like mental health assessments. By combining facial and gait analysis, this approach offers a more comprehensive view of human behavior, enabling the system to pick up on both emotional expressions and physical movement patterns. This dual approach greatly improves the chances of spotting subtle signs that might be missed if we only focused on one aspect. A key takeaway from this research is that mental health conditions can show up in both visible and behavioral ways. Facial expressions can indicate immediate emotional states, while gait patterns can reveal longer-term behavioral trends. By merging these two modalities, the system can cut down on uncertainty and boost prediction accuracy. This underscores the value of multimodal learning, where different data sources work together to create a fuller picture of the underlying condition.

One key point in this discussion is how well deep learning techniques can tackle complex data. Models like CNNs and LSTMs have proven to be quite effective at automatically learning high-level features, eliminating the need for manual feature engineering. When it comes to facial analysis, transfer learning really boosts performance by utilizing pre-trained knowledge, which is especially helpful when working with smaller datasets.

However, despite these advantages, the study uncovered several challenges. The system's performance heavily relies on the quality and diversity of the dataset. Factors like lighting conditions, camera angles, occlusions, and background noise can all impact facial analysis, while variations in walking environments, clothing, and physical conditions can affect gait analysis. These elements might restrict how well the model generalizes to real-world situations.

Moreover, the size of the dataset is crucial for the system's robustness. Smaller datasets can lead to overfitting, even with regularization techniques in play. While methods like cross-validation and data augmentation can help alleviate this problem, there's still a pressing need for larger and more varied datasets to enhance reliability. Additionally, cultural and individual differences in expressions and movement patterns pose challenges, as models trained on one group may not perform as well on another.

Ethical considerations also play a vital role in this conversation. Given that the system handles sensitive personal data, it's essential to address issues related to privacy, data security, and informed consent with great care. Implementing such systems should adhere to strict ethical guidelines to ensure that user data is safeguarded and used responsibly. Furthermore, the system should be viewed as a supportive tool for mental health professionals rather than a replacement.

CHAPTER 6

SUMMARY AND CONCLUSION

Detect AI-generated content and transform it into something that sounds more human with our AI Content Detector. Just paste your text, and you'll get accurate, natural-sounding results in no time! Here's the text we're looking at: The system uses a Convolutional Neural Network (CNN) to pull out facial features, paired with a recurrent model that analyzes how people walk. It also employs multimodal fusion techniques to boost the reliability of classifications. The main goal here is to see if this system can work effectively in real-life situations, rather than just creating a tool for clinical diagnostics. The findings show that facial cues deliver solid baseline performance, while gait features on their own have limited ability to distinguish between different cases, especially in messy video footage. However, when you combine both types of data, it really enhances prediction stability, minimizes class bias, and boosts the overall reliability of the model. Through various comparative analyses, including ROC curves, confusion matrices, and ablation studies, it's clear that multimodal models have the edge over unimodal ones, particularly when it comes to calibration and reducing overconfidence.

One key takeaway from the research is that multimodal systems tend to offer more conservative and explainable predictions, which is particularly beneficial for mental health screening applications. However, the study also points out some important limitations, such as the small size of the dataset, variability across different domains, and the significant influence of preprocessing quality, especially in gait analysis. Looking ahead, future efforts will aim to gather larger and more varied datasets, enhance preprocessing methods like pose estimation in less controlled environments, and delve into advanced techniques such as complex fusion strategies, self-supervised learning, and domain adaptation. It's also recommended to collaborate with domain experts and utilize longitudinal data to boost the clinical relevance and practical use of multimodal mental health detection systems.

This study presents a multimodal framework for mental health screening that leverages both facial expression analysis and gait-based behavioral indicators obtained from non-contact video inputs. The system is designed by integrating a Convolutional Neural Network (CNN) for extracting detailed facial features with a recurrent model, such as LSTM, for capturing temporal dynamics in gait patterns. These two modalities are combined using multimodal fusion techniques, which enhance the robustness and reliability of the classification process. The primary aim of the work is not to replace clinical diagnostic tools but to explore the feasibility and effectiveness of such automated systems in real-world, unconstrained environments. The experimental findings reveal that facial features serve as a strong baseline, providing significant discriminative power in identifying mental health conditions. In contrast, gait-based features, when used independently, show relatively lower performance, especially in unstructured and real-world video conditions where noise, occlusions, and variability are more prominent. However, when both modalities are integrated, the system demonstrates improved prediction stability, reduced bias toward specific classes, and better overall performance. This highlights the complementary nature of facial and gait cues, where each modality compensates for the limitations of the other.

Taking a closer look with evaluation techniques like ROC curves, confusion matrices, and ablation studies shows that multimodal models consistently have the upper hand over their unimodal counterparts. One standout benefit of the multimodal approach is its enhanced calibration, which means the model offers more trustworthy probability estimates instead of just overly confident predictions. This is especially crucial in mental health screening, where we prefer cautious and interpretable outputs over those that are highly confident but could be misleading. The system's knack for producing more conservative and explainable predictions makes it a better fit for real-world applications and supportive decision-making.

This work introduces an innovative framework for mental health screening that cleverly combines facial expression analysis with gait-based behavioral cues, all through non-contact video input. By utilizing a Convolutional Neural Network to extract facial features and a recurrent model to analyze gait dynamics, and then boosting performance with multimodal fusion techniques, the system showcases how artificial intelligence can be effectively applied to real-world mental health assessments. The findings reveal that facial cues provide a strong and dependable foundation for detection, while gait features on their own have limited ability to distinguish between conditions, especially in unstructured, real-world video scenarios. However, when both modalities are integrated, there's a notable improvement in prediction stability, a reduction in class bias, and an overall enhancement in classification reliability. Evaluation methods like ROC curves, confusion matrices, and ablation studies further validate that multimodal models surpass unimodal approaches, particularly regarding calibration and minimizing overconfidence.

One of the key takeaways from this study is that multimodal models tend to deliver more cautious and interpretable predictions. This is especially important in mental health screening, where being clear and reliable is crucial. Instead of functioning as a clinical diagnostic tool, the system we propose acts more like a supportive and initial screening aid. However, the study also points out some significant limitations, such as the small size of the dataset, variability across different domains, and the considerable impact of preprocessing quality, particularly in gait analysis. These issues suggest that we need to make further enhancements before this can be rolled out in real-world settings. The experimental results show that facial cues are incredibly effective as a baseline, providing a strong ability to identify mental health indicators. On the other hand, relying solely on gait-based features proves to be less effective, especially in real-world video situations where noise, variability, and environmental factors can affect performance. When we combine both facial and gait information, the system demonstrates better prediction stability, less class bias, and overall greater reliability. This highlights how facial and gait data complement each other, with each one helping to cover the weaknesses of the other.

6.1 Future Work

The proposed multimodal framework for detecting mental health issues shows a lot of promise, but there are still several areas that need more research and improvement to really boost its effectiveness and how well it works in the real world. One key direction for future work is to expand the dataset. Right now, the study is limited by a relatively small and not very diverse dataset, which impacts how well the model can generalize. Future efforts will aim to gather larger and more varied datasets that take into account differences in age, gender, cultural backgrounds, and environmental conditions. By including real-world variability, the system will be able to perform more reliably across different populations and settings.

Another crucial area for improvement is enhancing the preprocessing pipeline, especially when it comes to gait analysis. Since the gait features have shown limited ability to distinguish in unstructured video data, it's vital to improve techniques like pose estimation, silhouette extraction, and motion tracking in less controlled environments. Developing stronger preprocessing methods will help in pulling out cleaner and more meaningful gait features, which will ultimately enhance the overall performance of the system.

In the future, research will dive into more advanced multimodal fusion strategies. While the current system relies on standard fusion techniques, exploring more sophisticated methods like attention-based fusion and hybrid fusion models could enhance our understanding of the connections between facial and gait features. Plus, bringing in Self-Supervised Learning can tap into vast amounts of unlabeled data, which would lessen our reliance on annotated datasets and boost feature representation. Another important area to tackle is domain variability. The performance differences we see in various environments highlight the necessity for domain adaptation techniques. We can look into methods like transfer learning and domain generalization to ensure that the model maintains consistent performance across different conditions without needing extensive retraining.

Integrating longitudinal data—essentially, information gathered over time—can really enhance our understanding of how mental health conditions evolve. This approach allows us to go beyond just making static predictions; it enables ongoing monitoring and trend analysis, which is incredibly valuable for assessing mental health.

Working closely with mental health professionals is also key for future advancements. Their expertise can help fine-tune the outputs of our models, ensuring they are clinically relevant and easy to understand. This partnership can also steer the development of systems that better fit into real-world healthcare practices.

Looking ahead, we'll be focusing on making our models more explainable and transparent. By developing ways to interpret predictions clearly, we can boost trust and usability, especially in sensitive areas like mental health. We'll keep ethical considerations—like privacy, data security, and responsible AI use—at the forefront of our system development.

REFERENCES

- [1] R K. Kroenke, R. L. Spitzer, and J. B. Williams, "The PHQ-9: Validity of a brief depression severity measure," *J. Gen. Intern. Med.*, vol. 16, no. 9, pp. 606–613, 2001.
- [2] M. K. Al-Nafjan et al., "Detecting Depression Through Gait Data: Examining the Contribution of Gait Features," *Front. Psychiatry*, vol. 12, p. 628226, 2021.
- [3] S. R. P. et al., "Recognition of anxiety and depression using gait data recorded by Kinect," *Sci. Rep.*, vol. 15, Art. no. 06535, 2025. doi: 10.1038/s41598-025-06535-3.
- [4] H. Tang et al., "A gait assessment framework for depression detection using Kinect," *Proc. IEEE BIBM*, 2020. T. Lee et al., "Deep learning driven depression emotion recognition," *JMIR Mental Health*, 2023.
- [5] J. F. Cohn et al., "Facial expression as a marker of depression," *Psychol. Med.*, vol. 45, no. 6, pp. 1195–1205, 2015. L. Wang et al., "AI-assisted multimodal screening of depression," *Front. Psychiatry*, 2025.
- [6] S. Li and W. Deng, "Deep facial expression recognition: A survey," *IEEE Trans. Affect. Comput.*, vol. 11, no. 1, pp. 119–135, 2020. M. M. Ali et al., "Human pose estimation for clinical gait analysis," *JMIR Formative Research*, 2024.
- [7] T. Lee et al., "Deep learning driven depression emotion recognition," *JMIR Mental Health*, 2023.
- [8] Z. Dai et al., "Depression detection based on facial expression, audio and gait," *Proc. IEEE ICASSP*, 2023.
- [9] X. Li et al., "Multimodal depression recognition based on gait and rating scale," *Expert Syst. Appl.*, vol. 238, 2024.
- [10] L. Wang et al., "AI-assisted multimodal screening of depression," *Front. Psychiatry*, 2025.

- [11] C. Drummond and N. Holte, "C4.5, class imbalance, and cost sensitivity," SIGKDD Explorations, vol. 6, no. 1, pp. 25–38, 2005.
- [12] C. Lugaresi et al., "MediaPipe: A framework for building perception pipelines," arXiv:1906.08172, 2019.
- [13] M. M. Ali et al., "Human pose estimation for clinical gait analysis," JMIR Formative Research, 2024.
- [14] S. Du, Y. Tao, and A. M. Martinez, "Compound Facial Expressions of Emotion," PNAS, vol. 111, no. 15, pp. E1454–E1462, 2019.
- [15] P. Ekman and W. V. Friesen, Facial Action Coding System, 1978.
- [16] E. Seifert et al., "Detecting Depression Through Gait Data," Frontiers in Psychiatry, 2021.
- [17] M. S. Islam et al., "EmoPulse: Real-Time Mental Health via Facial Expressions and Psychometrics," 2024.

ANNEXURES

ANNEXURE 1

PLAGIARISM REPORT

ORIGINALITY REPORT			
6%	4%	5%	1%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS
PRIMARY SOURCES			
1	www.mdpi.com Internet Source	1%	
2	B. G. Nagaraja, S. Kannadhasan. "Information and Communication Systems", CRC Press, 2026 Publication	<1%	
3	Submitted to University of Witwatersrand Student Paper	<1%	
4	Jin Liu, Jianxin Wang, Yi Pan. "AI in MRI-based Brain Disease Prediction", CRC Press, 2026 Publication	<1%	
5	R. N. V. Jagan Mohan, B. H. V. S. Rama Krishnam Raju, V. Chandra Sekhar, T. V. K. P. Prasad. "Algorithms in Advanced Artificial Intelligence - Proceedings of International Conference on Algorithms in Advanced Artificial Intelligence (ICAAAI-2024)", CRC Press, 2025 Publication	<1%	
6	doctorpenguin.com Internet Source	<1%	
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8	ijcse.isroset.org Internet Source	<1%	
9	mljourney.com Internet Source	<1%	

ANNEXURE 2

JOURNAL DETAILS



ANNEXURE 3

SDGMAPPING

Project Title : **Facial And Gait Cuses For Mental Health Detection**

Table A1: Mapping with Sustainable Development Goals (SDGs)

SDG No.	SDG Title	Relevance Level	Justification
SDG 3	Good Health and Well-being	High	Enables early detection of mental health conditions like depression, improving overall well-being and reducing health risks.
SDG 4	Quality Education	Moderate	Supports research and awareness in mental health using advanced AI technologies, contributing to academic and practical learning.
SDG 9	Industry, Innovation and Infrastructure	High	Promotes innovation through AI, deep learning, and multimodal systems for healthcare applications.
SDG 10	Reduced Inequalities	Moderate	Provides accessible and low-cost mental health screening, helping underserved populations.
SDG 11	Sustainable Cities and Communities	Moderate	Encourages healthier communities by improving mental health monitoring in urban environments.
SDG 13	Climate Action	Low	Indirectly contributes by promoting digital healthcare solutions, reducing the need for physical travel and resources.

The proposed project on *Facial and Gait Cues for Mental Health Detection* is closely aligned with several Sustainable Development Goals (SDGs) through its focus on AI-based mental health screening. It strongly contributes to SDG 3: Good Health and Well-being, as the system enables early detection of depression using facial expressions and gait patterns, thereby supporting timely intervention and improving overall mental health outcomes. The project also supports SDG 4: Quality Education by enhancing research and awareness in the field of artificial intelligence and mental health, contributing to both academic learning and practical applications.

Furthermore, the research aligns with SDG 9: Industry, Innovation and Infrastructure by utilizing advanced technologies such as Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), and multimodal fusion techniques to develop an innovative healthcare solution. It contributes to SDG 10: Reduced Inequalities by providing a non-contact, cost-effective, and accessible mental health screening system that can be deployed using standard devices like smartphones, making it beneficial for underserved populations.

In addition, the project supports SDG 11: Sustainable Cities and Communities by promoting healthier communities through scalable and remote mental health monitoring solutions. Lastly, it indirectly relates to SDG 17: Partnerships for the Goals, as the successful implementation and real-world deployment of such systems require collaboration between researchers, healthcare professionals, and technological institutions. Overall, the project demonstrates how integrating AI with healthcare can address critical mental health challenges while contributing to global sustainable development objectives.

ANNEXURE 4

PO– PSO MAPPING

TITLE: FACIAL AND GAIT CUES FOR MENTAL HEALTH DETECTION

PO Mapping Table

PO / PSO	Mapping Level	Justification
PO1	3	Application of machine learning, CNN, and LSTM concepts for mental health detection
PO2	3	Analysis of behavioral patterns like facial expressions and gait signals
PO3	3	Design and development of a multimodal AI-based detection system
PO4	2	Experimental validation using datasets, ROC curves, and confusion matrix
PO5	3	Use of modern tools such as Python, TensorFlow, Media Pipe, OpenCV
PO6	3	Addresses societal issue of mental health and early depression screening
PO7	2	Promotes sustainable healthcare solutions through technology
PO8	3	Ensures ethical handling of sensitive biometric data (face & gait)

PO9	2	Team collaboration in research, development, and testing
PO10	2	Documentation, report writing, and presentation of results
PO11	1	Basic project planning and execution
PO12	3	Learning emerging technologies in AI, deep learning, and healthcare analytics

PSO Mapping Table

PSO	Mapping Level	Justification
PSO1	3	Application of machine learning for knowledge extraction from behavioral data
PSO2	3	Development of socially relevant AI-based healthcare solution
PSO3	3	Design and implementation of a real-time mental health detection system

ANNEXURE 5

ADDITIONAL SCREENSHOTS

